

A Project Report on
Weather Forecast Application

By:

Raj Dangui (1292240011)

Aditya Bhagat (1292240016)

Kartik Kachwal (1292240020)

MCA – I, SEM-II

Batch 2024-2026



Dr. Vishwanath Karad MIT

World Peace University, Pune

Pune- 411038

In Partial Fulfillment of the Degree of Master in Computer Application (M. C. A.)

Under The Guidance Of

Prof.Gajanan Garkar

Academic Year 2024-25

Total Number of Tables

No.	Table Name	Description
1	users	Stores user authentication details
2	locations	Stores user-saved locations
3	weather	Stores current weather data
4	forecast_weather	Stores forecast weather details
5	alerts	Stores weather alerts and in-app notifications
6	news	Stores weather-related news articles
7	sun_moon_info	Stores sunrise, sunset, and moon phase information

Database Tables and Schema

Table: users (Stores User Authentication Details)

Primary Key: user_id

Constraints: email must be unique and not null.

Data Schema

Column Name	Data Type	Constraints
user_id	INTEGER	PRIMARY KEY AUTOINCREMENT
name	TEXT	NOT NULL
email	TEXT	UNIQUE, NOT NULL
password	TEXT	NOT NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

Table: locations (Stores User-Saved Locations)

Primary Key: location_id

Foreign Key: user_id references users(user_id), deletes related locations if the user is deleted.

Data Schema

Column Name	Data Type	Constraints
location_id	INTEGER	PRIMARY KEY AUTOINCREMENT
user_id	INTEGER	FOREIGN KEY REFERENCES users(user_id) ON DELETE CASCADE
city_name	TEXT	NOT NULL
latitude	REAL	NOT NULL
longitude	REAL	NOT NULL

Table: weather (Stores Current Weather Data)

Primary Key: weather_id

Foreign Key: location_id references locations(location_id).

Data Schema

Column Name	Data Type	Constraints
weather_id	INTEGER	PRIMARY KEY AUTOINCREMENT
location_id	INTEGER	FOREIGN KEY REFERENCES locations(location_id)
temperature	REAL	NOT NULL
humidity	INTEGER	NOT NULL
wind_speed	REAL	NOT NULL
uv_index	REAL	NOT NULL
air_quality	TEXT	NOT NULL
sunrise_time	TEXT	NOT NULL
sunset_time	TEXT	NOT NULL
weather_desc	TEXT	NOT NULL
timestamp	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

Table: forecast_weather (Stores Forecast Weather Data for Upcoming Days)

Primary Key: forecast_id

Foreign Key: location_id references locations(location_id).

Data Schema

Column Name	Data Type	Constraints
forecast_id	INTEGER	PRIMARY KEY AUTOINCREMENT
location_id	INTEGER	FOREIGN KEY REFERENCES locations(location_id)
date	TEXT	NOT NULL
temperature	REAL	NOT NULL
humidity	INTEGER	NOT NULL
wind_speed	REAL	NOT NULL
uv_index	REAL	NOT NULL
air_quality	TEXT	NOT NULL
sunrise_time	TEXT	NOT NULL
sunset_time	TEXT	NOT NULL
weather_desc	TEXT	NOT NULL

Table: alerts (Stores Weather Alerts and In-App Notifications)

Primary Key: alert_id

Foreign Key: forecast_id references forecast_weather(forecast_id).

Data Schema

Column Name	Data Type	Constraints
alert_id	INTEGER	PRIMARY KEY AUTOINCREMENT
forecast_id	INTEGER	FOREIGN KEY REFERENCES forecast_weather(forecast_id)
user_id	INTEGER	FOREIGN KEY REFERENCES users(user_id) (NULL if general notification)
title	TEXT	NOT NULL
message	TEXT	NOT NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
is_read	BOOLEAN	DEFAULT 0 (Unread by default)

Table: news (Stores Weather News and Updates)

Primary Key: news_id

Data Schema

Column Name	Data Type	Constraints
news_id	INTEGER	PRIMARY KEY AUTOINCREMENT
title	TEXT	NOT NULL
content	TEXT	NOT NULL
published_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

Table: sun_moon_info (Stores Sun & Moon Timings)

Primary Key: info_id

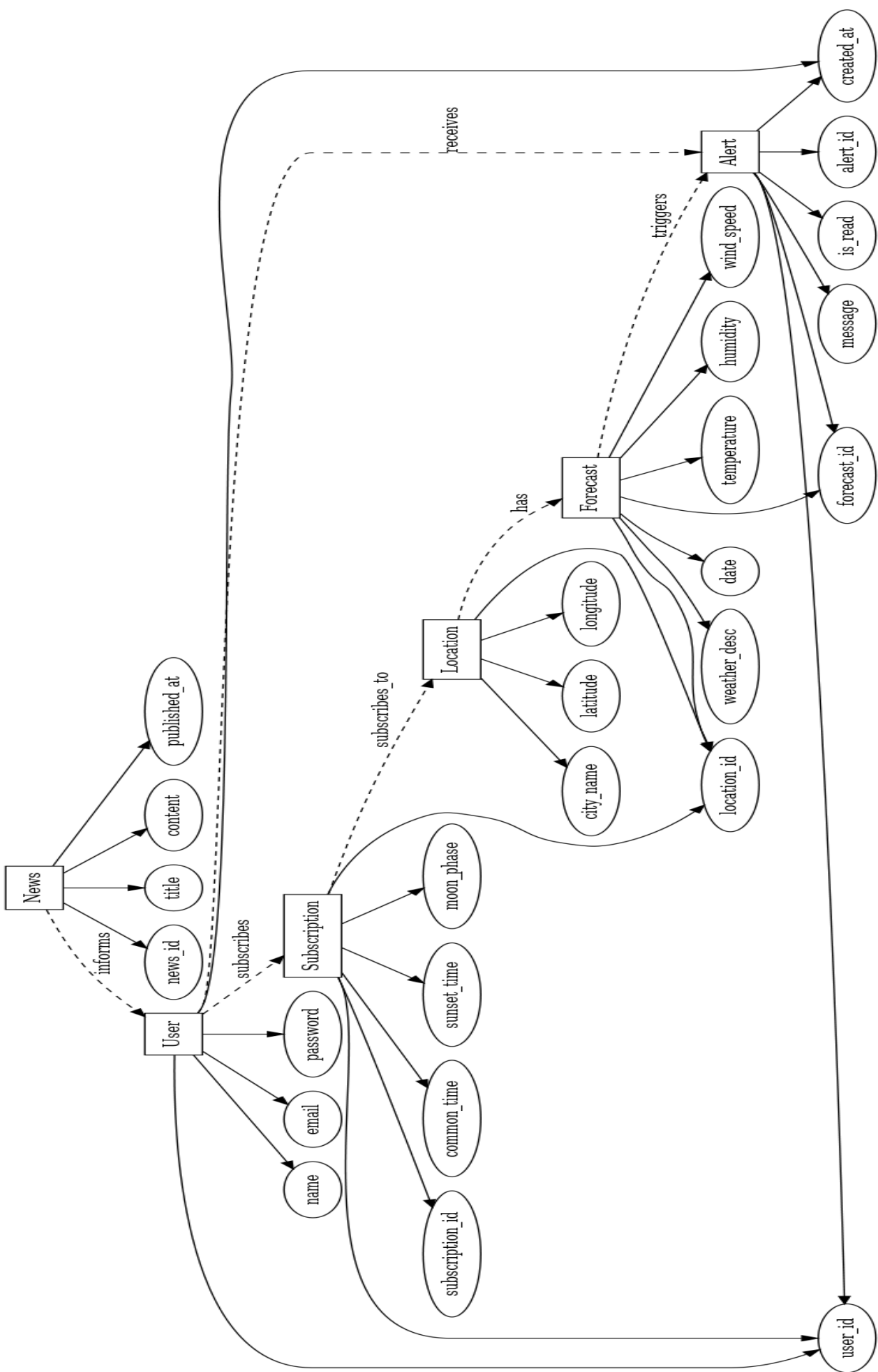
Foreign Key: location_id references locations(location_id).

Data Schema

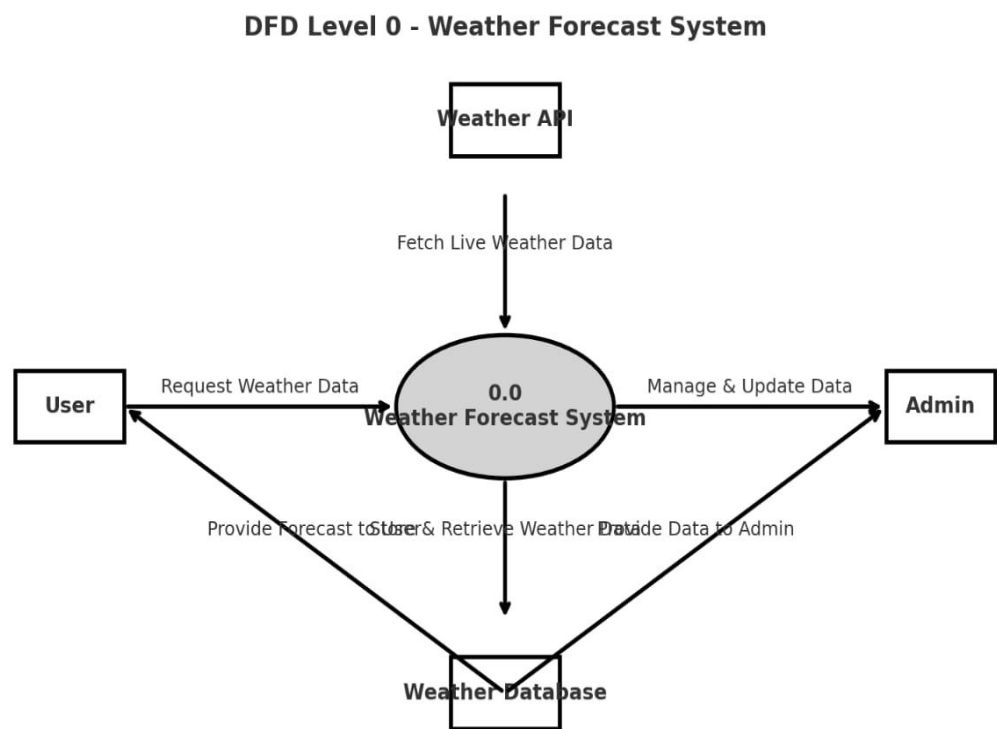
Column Name	Data Type	Constraints
info_id	INTEGER	PRIMARY KEY AUTOINCREMENT
location_id	INTEGER	FOREIGN KEY REFERENCES locations(location_id)
date	TEXT	NOT NULL
sunrise_time	TEXT	NOT NULL
sunset_time	TEXT	NOT NULL
moon_phase	TEXT	NOT NULL

Database Diagrams

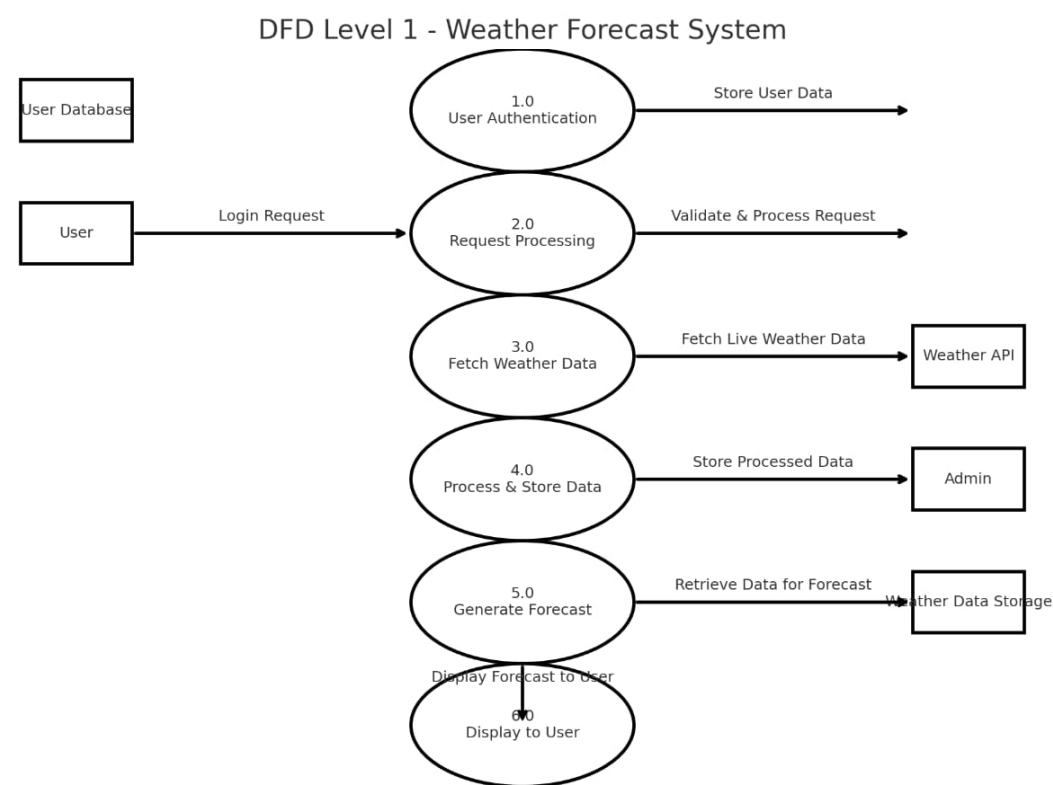
1. Entity-Relationship (ER) Diagram



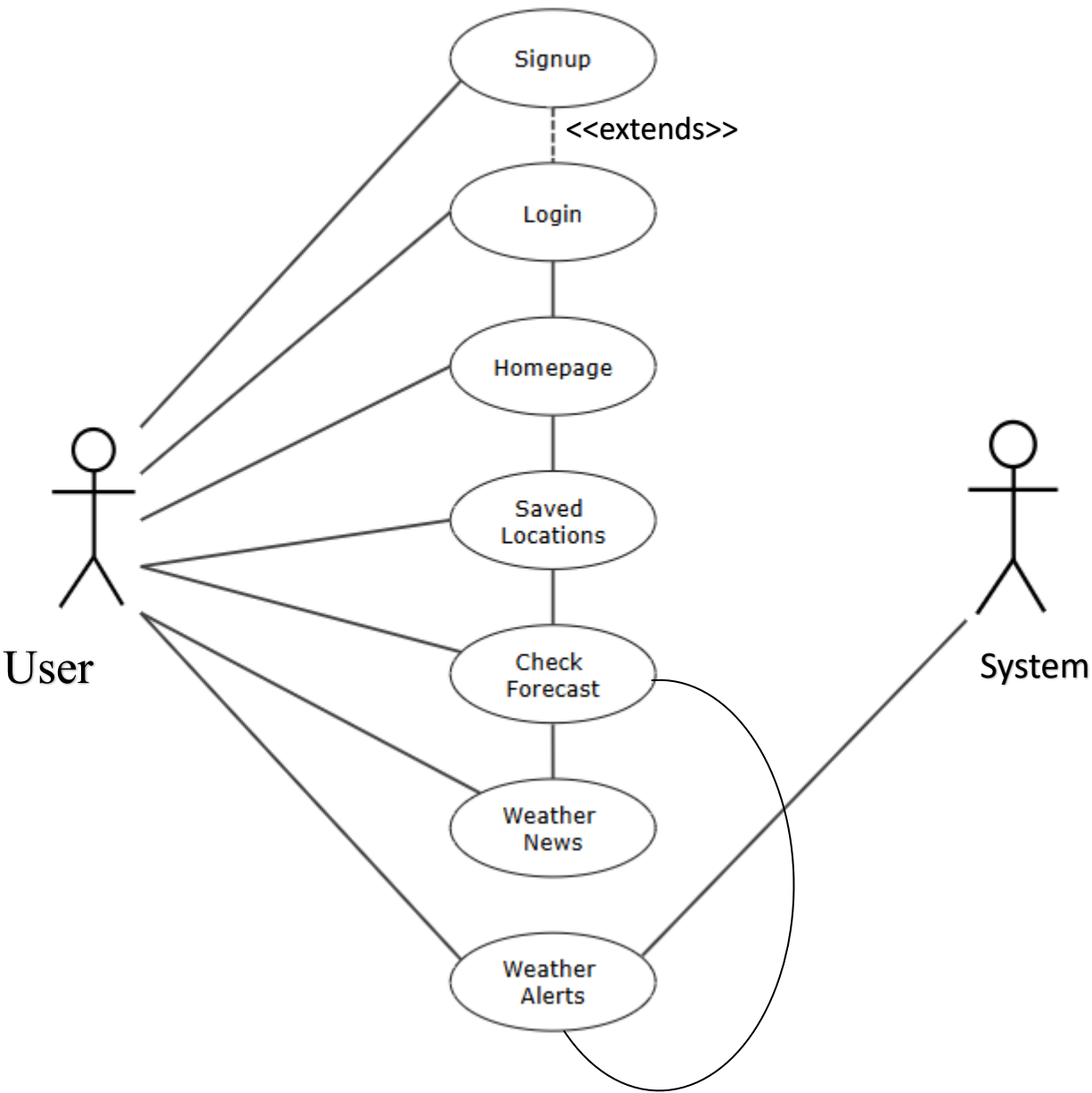
2. Data Flow Diagram (DFD) - Level 0



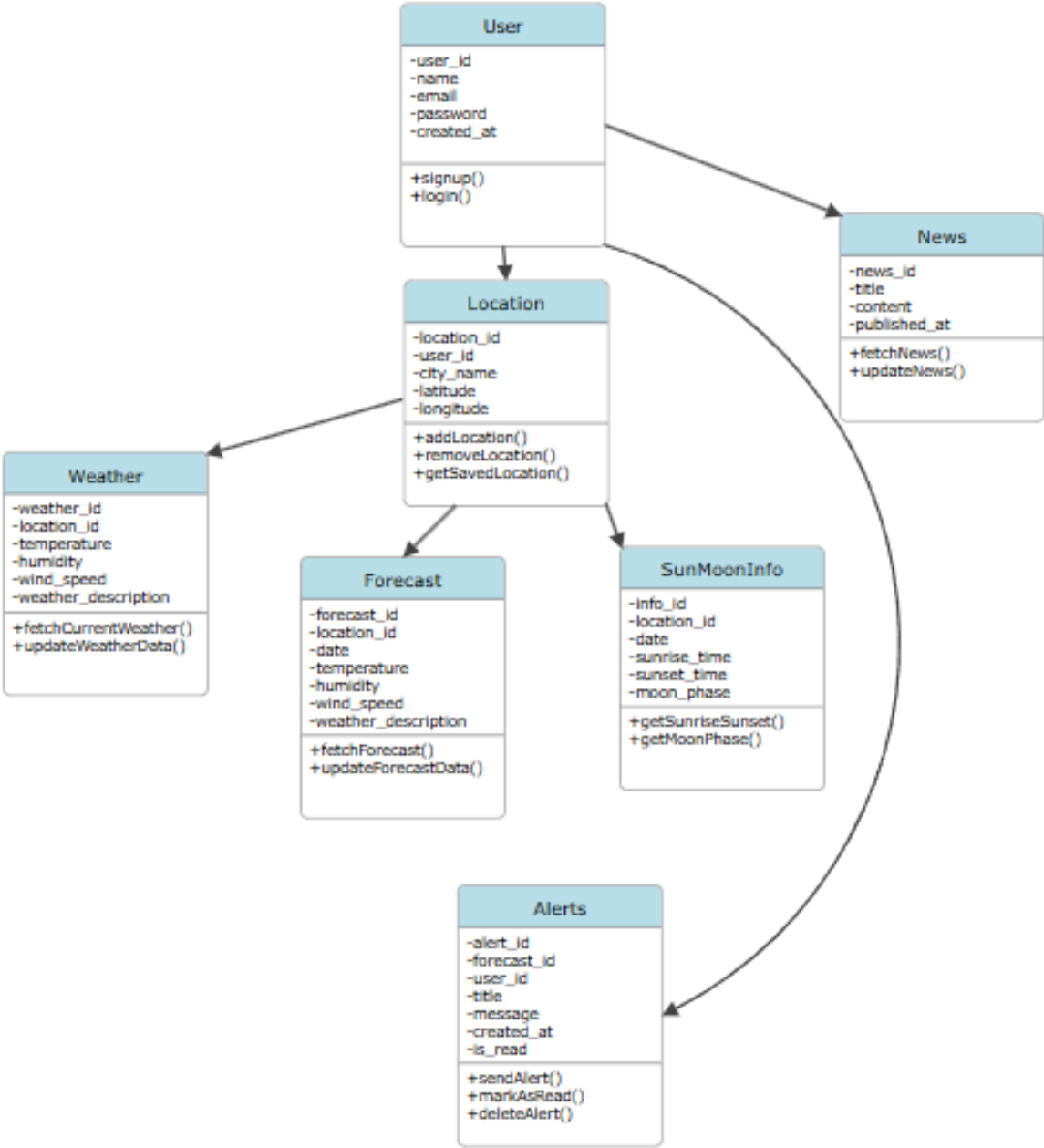
3. Data Flow Diagram (DFD) - Level 1



4. UML Use Case Diagram



5. UML Class Diagram



6. Sequence Diagram

